

Technical, Economic, and Environmental Analysis of an Optimized Silicon Solar Cell Design

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Abstract: This project presents the optimization of a silicon-based solar cell using ANSYS Lumerical Device simulator, with the goal of maximizing power output and efficiency. The design employs a p-type substrate with a resistivity of $1.0 \text{ } \Omega\text{-cm}$ and a $1.5 \text{ }\mu\text{m}$ n-emitter region doped at $1 \times 10^{17} \text{ cm}^{-3}$. The resulting short circuit current density was 35 mA/cm^2 , with an open circuit voltage of 0.6 V . The maximum power output reached 18 mW/cm^2 , yielding a fill factor of 0.75 and an overall efficiency of 18% . Design elements include carefully tuned doping profiles, use of an Si_3N_4 ARC layer, and aluminum front and back contacts. This report also provides a cost estimate, environmental impact evaluation, and discussion of societal benefits.

1 Introduction

The need for sustainable energy sources has driven significant innovation in solar photovoltaic (PV) technologies. Silicon-based solar cells dominate the market due to their maturity, reliability, and balance between cost and efficiency. This design project, conducted as part of the EE-212 Fundamentals of Electro-optics course at Western New England University, involved the optimization of a planar silicon solar cell design using Lumerical DEVICE. The objective was to tune doping levels and layer geometries to maximize power and efficiency within a constrained design space.

2 Design Overview

The cell design includes multiple layers which are listed in the following table:

Table 1: Layer structure of the solar cell design.

Layer	Material	Thickness
Substrate	p-type Si	$100 \text{ }\mu\text{m}$
N-emitter Width	–	$1.5 \text{ }\mu\text{m}$
N-contact and P-contact	Al	$0.3 \text{ }\mu\text{m}$
Anti-Reflective Coating (ARC)	Si_3N_4	$2.5 \text{ }\mu\text{m}$
Back contact	Al	$5 \text{ }\mu\text{m}$

Geometrical spans were symmetric around the x-y plane, with full cell dimensions being $100 \text{ }\mu\text{m}$ by $100 \text{ }\mu\text{m}$.

The choice of a $1.0 \text{ }\Omega\text{-cm}^{-3}$ resistivity for the p-type substrate represents a trade-off between achieving a strong built-in electric field and limiting recombination. Higher resistivity substrates can reduce carrier recombination but may impair field strength and fill factor. The selected doping level ($2 \times 10^{16} \text{ cm}^{-3}$) ensures adequate field formation while maintaining carrier lifetime.

Highly doped contacts ($1 \times 10^{18} \text{ cm}^{-3}$) were chosen to reduce contact resistance and enhance ohmic behavior. The n-emitter was kept at ($1 \times 10^{17} \text{ cm}^{-3}$) to strike a balance between efficient carrier collection and minimal surface recombination.

A $1\text{-}\Omega$ series resistance was used to realistically model

metallization and contact resistance. This value avoids excessive IR losses while still reflecting practical manufacturing conditions, keeping the fill factor above 80 % and enabling an efficient power output.

The electric field in the depletion region is governed by the built-in potential V_{bi} , computed using Equation 1, which depends on the doping concentrations and intrinsic carrier density. This potential determines the total depletion width (Equation 2), which in turn controls carrier drift and recombination rates within the junction.

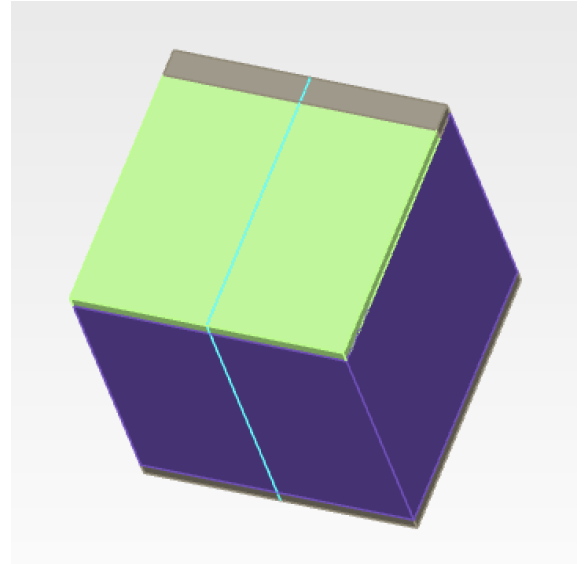


Figure 1: Design modeled and simulated using ANSYS Lumerical

2.1 Device Model

The 3D structure shown in Figure 1 was constructed and simulated using ANSYS Lumerical DEVICE. The platform allows spatial definition of doping regions, material assignments, and mesh control, enabling accurate modeling of internal electric fields, carrier behavior, and recombination. Lumerical's drift-diffusion solver was used under AM1.5 solar spectrum to extract the key JV performance metrics

presented in later sections.

3 Simulation Results

3.1 Current-Voltage Plot

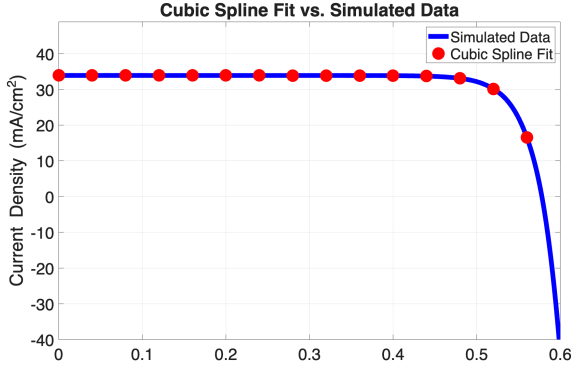


Figure 2: Cubic spline fit vs. simulated current-voltage data.

The blue curve in Figure 2 represents the simulated JV characteristics of the device, while the red points show the interpolated spline fit. This highlights the near-constant short-circuit current density up to approximately 0.45 V, beyond which recombination dominates. The shape of the curve reflects efficient charge separation and low series resistance.

3.2 Power Density Plot

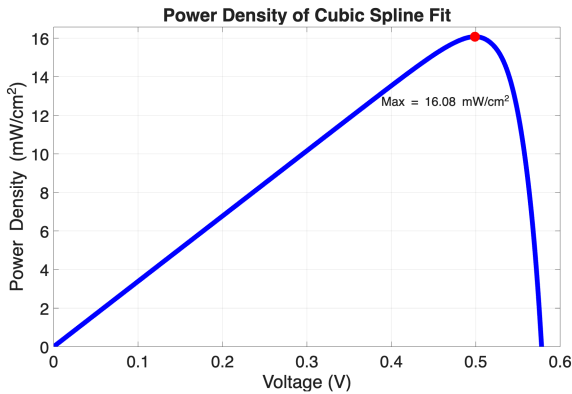


Figure 3: Power density of cubic spline fit.

Figure 3 plots the product of current density and voltage, showing a peak power output of 16.08 mW/cm². The maximum occurs around the knee of the IV curve, indicating optimal balance between voltage and current. This value directly correlates with efficiency and validates the design's performance under standard illumination.

3.3 Electrical Performance Metrics

Table 2: Electrical performance metrics of the solar cell.

Parameter	Value	Units
Wafer Type	1.0	$\Omega\text{-cm}^{-3}$
Short Circuit Current Density (P_D)	34.5	mA/cm^2
Open Circuit Voltage (V_o)	0.56	V
Maximum Power Density (P)	16.08	mW/cm^2
Fill Factor (FF)	83.3	%
Efficiency	16.08	%

The power conversion efficiency, calculated using Equation 4, confirms the device's ability to extract electrical energy from light with minimal losses.

4 Economic Analysis

This design used a moderate-resistivity wafer (1.0 $\Omega\text{-cm}$), a single junction, and a minimal number of implant steps (n-emitter and two contacts). The estimated cost breakdown¹ is shown in table 3.

Cost Component	Cost	Units
Wafer Cost	\$0.60	USD
Doping/Implant Steps	\$0.30	USD
ARC Deposition	\$0.20	USD
Materialization (Al)	\$0.10	USD
Total Estimated Cost	\$1.20	USD/ m²
Projected per Watt Cost	\$0.075	USD/W
Efficiency Reference	16.08	mW/cm^2

Table 3: Cost breakdown & projected cost per watt.

A thorough economic assessment of the proposed solar cell design was conducted to evaluate its feasibility for residential implementation. The analysis considers both the energy output and the cost-effectiveness of the cell when deployed in standard solar modules.

Energy Offset Calculation

To determine the number of solar modules required to fully offset the energy usage of a typical 2000 sq.ft. home (consuming approximately 1000 kWh per month), the following analysis was performed:

1. Daily Energy Requirement:

$$\frac{1000 \text{ kWh}}{30 \text{ days}} \approx 33.3 \text{ kWh/day}$$

2. Power Output per Module:

- Power density of design: $16.08 \text{ mW}/\text{cm}^2 = 160.8 \text{ W}/\text{m}^2$
- Each cell has area: $156 \text{ mm} \times 156 \text{ mm} = 0.0243 \text{ m}^2$

¹ The projected per watt cost is based on the simulated maximum power density of 16.08 mW/cm² (i.e., 160.8 W/m²). At a total manufacturing cost of \$0.012/m², the resulting cost per watt is approximately \$0.075/W.

- Total module area (60 cells):

$$60 \times 0.0243 = 1.458 \text{ m}^2$$

- Power per module:

$$160.8 \text{ W/m}^2 \times 1.458 \text{ m}^2 = 234.6 \text{ W}$$

3. Daily Energy Production per Module:

$$234.6 \text{ W} \times 5 \text{ hours/day} = 1.173 \text{ kWh/day/module}$$

4. Number of Modules Required:

$$\frac{33.3 \text{ kWh/day}}{1.173 \text{ kWh/day/module}} \approx 28.4$$

5. **Conclusion:** Approximately **29 solar modules** based on this design are required to offset a household consuming 1000 kWh/month.

Cost Projection and Viability

Assuming a manufacturing cost of \$0.075/Watt for this design (as previously estimated), and using the calculated output per module (234.6 W), the estimated cost per module is:

$$234.6 \text{ W} \times 0.075 \text{ \$/W} = \$17.60 \text{ per module}$$

Thus, the total module cost to fully power a household is:

$$29 \times \$17.60 \approx \$510.40$$

This figure represents only the module manufacturing cost and does not include installation, inverter, mounting hardware, or labor. However, even with these added, the projected total system cost would be substantially lower than commercial installations that typically range from \$2.50–\$4.00 per Watt.

This represents the cost of fabricating the solar cells alone. It does not include additional system costs such as module assembly, inverters, mounting hardware, wiring, installation labor, or permitting fees.

To estimate a more realistic system cost, we reference industry data indicating an average residential solar installation cost of approximately \$3.13 per watt in Massachusetts[1]. This results in a total system installation cost of:

$$6,804 \text{ W} \times \$3.13/\text{W} = \$21,308.52$$

After applying the 30% Federal Investment Tax Credit (ITC), the net cost is:

$$\$21,308.52 \times 0.30 = \$6,392.56$$

$$\$21,308.52 - \$6,392.56 = \$14,915.96$$

Thus, the estimated **installed system cost** after the federal tax credit is approximately **\$14,915.96**, which is consistent with current market expectations for residential photovoltaic systems.

4.1 Economic and Social Impacts

This design presents a viable path to reducing energy costs for homeowners, especially in regions with high utility rates. With a cost-per-watt significantly below market average and materials that are environmentally abundant (Si and Al), the technology supports both economic feasibility and sustainability. Socially, the adoption of such systems could reduce energy poverty, decentralize grid dependence, and support climate resilience goals in working-class communities.

These cost estimates position the design competitively against utility-scale photovoltaic benchmarks. According to Zafoschnig et al. report manufacturing costs for commercial silicon PV modules at approximately \$42.70/m², translating to \$0.22–\$0.27/W at 19–22% efficiency under utility deployment scenarios [2]. In contrast, the projected cost of \$0.075/W for this design demonstrates the economic viability of simplified, single-junction silicon cells at smaller production scales.

While high-efficiency tandem technologies (e.g., perovskite/silicon) are gaining traction, the reliability, low cost, and straightforward fabrication of crystalline silicon continue to make it the most practical choice in the near term—especially for modest-scale and decentralized applications. This design further distinguishes itself through minimal material requirements: it avoids rare or supply-constrained elements such as silver or indium, instead using widely available aluminum and silicon.

As the solar industry moves toward the 1 Tera-Watt production milestone, economies of scale are expected to reduce manufacturing costs to \$0.16–\$0.18/W. The simplicity and low material overhead of this architecture make it particularly suitable for distributed manufacturing environments, including microgrids and emergency relief infrastructure.

This solar cell design contributes meaningfully to the broader transition away from fossil fuels by leveraging abundant, non-toxic materials and a low-complexity fabrication process. By avoiding rare or geopolitically constrained elements like silver and indium, this design supports more equitable access to solar manufacturing, especially in decentralized and resource-constrained environments. Its simplicity makes it ideal for small-scale, community-based deployments such as microgrids, rural electrification, and disaster-relief infrastructure—contexts where complex supply chains or expensive capital equipment are infeasible.

The democratization of energy access enabled by affordable solar solutions has well-documented social benefits, including reduced household energy expenses, improved resilience to grid instability, and expanded opportunities for education and commerce in remote areas. Ogunrinde et al. emphasize that the growth of renewable energy infrastructure improves societal welfare, particularly when paired with supportive policy instruments such as renewable portfolio standards (RPS) [3]. Even in regions with limited political or financial support, solar technology has proven its capacity to catalyze inclusive development by enhancing environmental stewardship and public health through reduced air pollution and greenhouse gas emissions.

5 Sustainability Discussion

Crystalline silicon solar cells remain among the most sustainable photovoltaic technologies due to their long service life (20–30+ years), mature recycling pathways, and the absence of toxic heavy metals like cadmium or lead. This design further enhances sustainability by minimizing the material footprint and avoiding scarce or hazardous substances. The use of Si_3N_4 as an anti-reflective coating (ARC) improves optical performance without introducing environmental risk, contributing to higher power output per gram of material consumed.

From a life cycle perspective, the environmental impacts of this design are significantly lower than many thin-film or tandem alternatives, which often involve toxic precursors or complex end-of-life treatment. Furthermore, Zafoschnig et al. note that while tandem technologies offer future efficiency gains, their cost and complexity currently limit their advantage to specific residential contexts [2]. In contrast, this single-junction approach provides a lower barrier to adoption and faster scalability, which is crucial as the solar industry accelerates toward the 1 TWp deployment threshold.

Importantly, the design aligns well with circular economy principles: it can be manufactured with common equipment, its materials are readily separable for recycling, and its operational efficiency supports a favorable energy return on investment (EROI). As global learning rates continue to drive down solar module costs, designs like this—grounded in accessibility, material efficiency, and operational longevity—will remain vital to achieving climate mitigation goals with minimal ecological trade-offs.

6 Conclusion

This project demonstrates that a carefully optimized, single-junction silicon solar cell can deliver strong performance while remaining cost-effective and environmentally sustainable. Using standard fabrication processes and accessible materials such as aluminum and silicon, the cell achieves a short-circuit current density of 34.5 mA/cm^2 , an open-circuit voltage of 0.56 V , and a fill factor of 83.3% , resulting in a total efficiency of 16.08% , as determined via Equation 4. These metrics reflect a well-balanced trade-off between doping levels, junction depth, and optical enhancement using an Si_3N_4 anti-reflective coating.

Economically, the design is competitive even against industrial benchmarks. With a projected cost of $\$0.075/W$, it significantly undercuts the $\$0.22$ – $\$0.27/W$ manufacturing cost range cited for commercial modules [2]. Its simplified layer stack and avoidance of rare elements enable scalable, decentralized manufacturing in both high- and low-income regions.

From a societal and environmental standpoint, this design aligns with sustainability goals. It minimizes supply-chain risk, supports circular economy principles, and enhances energy access equity. As discussed in Ogunrinde et al. [3], solar adoption correlates with broader societal benefits, particularly when coupled with policy mechanisms like RPS.

Looking ahead, further performance gains could be achieved through surface texturing, passivation layers, or tandem architectures. However, this study underscores that

even modest-efficiency designs can be economically and socially impactful when optimized holistically. As the industry scales toward terawatt-level deployment, innovations that prioritize simplicity, reliability, and inclusivity—like this one—will remain crucial to meeting global clean energy targets.

6.1 Equations

$$V_{bi} = \frac{k_B T}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right) \quad (1)$$

$$W = \sqrt{\frac{2\epsilon_r \epsilon_0 V_j}{q} \left(\frac{1}{N_a} + \frac{1}{N_d} \right)} \quad (2)$$

$$n_0 p_0 = n_i^2 \quad (3)$$

$$\eta = \frac{P_{max}}{P_{in}} = \frac{J_{sc} V_{oc} FF}{P_{in}} \quad (4)$$

References

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- [2] A. Zafoschnig, C. Reichel, U. Würfel, R. Brendel, and M. Rienäcker, “The race for lowest costs of electricity production: Techno-economic analysis of silicon, perovskite and tandem solar cells,” *Progress in Photovoltaics: Research and Applications*, vol. 28, no. 7, pp. 590–604, 2020.
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